

Open-Source Cross-Domain Datasets Enabled by LLM-Driven Digital Twins and Network Telemetry

SEN SHEN^{†,1}, WANXIN ZHAO^{†,1}, XUEWEN CHEN^{†,1}, JING HAN¹, XUEQING ZHOU¹, RUIZHI YANG¹, PEIYUN GE¹, ZHUOJUN CAI¹, VAIGAI YOKAR¹, YULEI WU¹, SHUANGYI YAN^{*1}, AND DIMITRA SIMEONIDOU¹

¹Smart Internet Lab, Faculty of Science and Engineering, University of Bristol, Bristol, United Kingdom.

[†]These authors contributed equally to this work.

^{*}shuangyi.yan@bristol.ac.uk

Compiled September 15, 2026

Machine-learning-enabled network research requires both practical telemetry and controllable synthetic data, but optical transport, fibre sensing, and wireless access data are usually collected and released separately. We present an openly available cross-domain dataset framework with two complementary parts: independently acquired practical datasets organised through common metadata, and generated datasets produced through an LLM-driven multi-agent digital-twin (DT) workflow. “Cross-domain” denotes common representation, repository, and task-level fusion; it does not imply contemporaneous end-to-end measurement. On a constructed optical–wireless interoperability benchmark with chronological splitting, a three-dimensional privacy-aware wireless representation achieves binary service-risk Macro-F1 0.646 ± 0.005 while moderately reducing observed CPE-attribute leakage relative to raw telemetry. In the optical generation case, 8,192 GNPY execution records correspond to 6,144 unique configurations, exactly matching an independent deterministic enumerator. A 96-case replay establishes historical numerical compatibility and quantifies sensitivity to a current GNPY version. The LLM workflow is therefore presented as a feasibility demonstration, while the semantic module is evaluated as an empirical privacy–utility trade-off rather than a privacy guarantee.

1. INTRODUCTION

In recent years, machine learning (ML) has been widely investigated in telecommunication networks to support intelligent and autonomous network operation. In optical networks, ML has been applied to quality of transmission (QoT) estimation, traffic prediction, anomaly detection, and resource allocation [1]. In wireless networks, ML-based methods have also been explored for radio resource management, multi-access optimisation, traffic prediction, mobility management, and service quality assurance [2]. These advances indicate a clear trend towards AI-native network intelligence, where data-driven models support monitoring, optimisation, and closed-loop control.

However, the effectiveness of ML models strongly depends on representative data. Operational network records can expose topology, device configurations, traffic, and service conditions, which limits public release. Existing optical and wireless datasets [3, 4] also tend to address a single domain, task, or campaign. This restricts studies of services whose behaviour

depends on both wireless access and optical transport. Heterogeneous infrastructures, telemetry interfaces, data formats, time scales, and privacy requirements make such datasets difficult to organise and combine. Synthetic data can complement practical measurements by covering controllable or rare conditions, although DT-based generation still requires substantial effort to specify scenarios, execute simulations, and validate the results.

To address these challenges, this paper presents an openly available cross-domain dataset framework that supports both practical and generated datasets: practical data are organised through network telemetry and semantic abstraction, whereas synthetic data are demonstrated through an LLM-driven multi-agent DT workflow. The term cross-domain has two precise meanings in this work. First, independently acquired optical-transport, fibre-sensing, and wireless-access datasets are exposed through a common repository and metadata structure. Second, a constructed interoperability benchmark combines optical and wireless feature sequences to test whether compact wireless representations remain useful to a common downstream

model. Because the practical campaigns were conducted at different times, this benchmark does not represent simultaneous end-to-end measurements and is not used to infer a physical causal relationship between the domains.

The main contributions are as follows. (1) We consolidate practical optical-transport, fibre-sensing, and multi-access wireless datasets with their acquisition context, formats, sampling information, and repository access terms. The telemetry platform builds on our previous implementations [5, 6]; the present contribution is their dataset-level organisation and cross-domain interface. (2) We specify and evaluate hand-crafted and learned wireless semantic representations using chronological partitions, train-only preprocessing, repeated runs, conventional baselines, and multiple privacy attackers. (3) We demonstrate an LLM-driven multi-agent interface to GNPpy and ns-3 DT engines and compare its optical configuration set with an independent deterministic enumerator. (4) We assess simulator provenance and replayability with historical-compatibility and current-version GNPpy checks, separating workflow feasibility, numerical reproducibility, and physical calibration.

The remaining sections are organised as follows. Section II reviews cross-domain data, semantic abstraction, DTs, and LLM-driven network automation. Section III presents the unified framework, Section IV describes the telemetry architecture and practical datasets, Section V presents the LLM-driven multi-agent DT demonstration, and Section VI concludes the paper.

2. RELATED WORK

A. Cross-Domain Data Availability and Network Telemetry

With the increasing use of ML in telecommunication networks, data have become a key enabling resource for model training, validation, inference, and updating. In cross-domain optical-wireless networks, practical telemetry is particularly valuable because it captures communication performance, device states, service requests, and environmental variations under real operating conditions. In optical transport and fibre-sensing systems, such data may include bit error rate (BER), optical signal-to-noise ratio (OSNR), Q-factor, optical power, route information, configuration records, device alarms, state-of-polarisation (SOP) evolution, backscatter traces, distributed acoustic sensing signals, and vibration-related features [7–9]. In wireless access networks, standardised telemetry may include throughput, packet loss, delay, signal strength, cell load, mobility, traffic distribution, and per-device access statistics [10, 11]. The value of these heterogeneous data lies not only in individual measurements, but also in their temporal alignment and shared service context across optical transport, fibre sensing, and wireless access domains.

However, practical cross-domain data are difficult to collect, align, and share. Optical transport, fibre sensing, and wireless access systems rely on different infrastructures, telemetry interfaces, data formats, and time scales. In addition, operational data may expose real network topology, configuration, device states, customer traffic, service conditions, and operator policies, thereby limiting public data sharing. Consequently, only a limited number of open optical-network datasets are available. Representative examples include the Microsoft wide-area optical backbone dataset for operational network-performance analysis [4, 12], the Fraunhofer HHI QoT dataset collection for ML-assisted QoT estimation in elastic optical networks [13], and the Alibaba disaggregated production-network dataset for field-data-based QoT estimation and parameter opti-

timisation [3, 14]. Recent dataset-oriented efforts have further covered space-division-multiplexing networks, fibre sensing, optical-device modelling, and radio-over-fibre systems [15–18]. Nevertheless, most existing datasets remain limited to a single domain, topology, task, or measurement campaign, and rarely provide time-aligned optical-transport, fibre-sensing, and wireless-access observations under a common service context.

Network telemetry provides the basis for scalable data collection [19]. Model-driven and streaming telemetry solutions based on NETCONF/YANG, gRPC, gNMI, and publish–subscribe mechanisms have been developed to expose network states and performance measurements [19, 20]. In optical networks, prior studies have demonstrated telemetry services for real-time monitoring in disaggregated optical networks, streaming-telemetry-assisted diagnosis, Kafka-based optical monitoring frameworks, and unified monitoring platforms integrating telemetry collection, time-series storage, configuration databases, and streaming pipelines [5, 21–23]. In wireless networks, O-RAN and AI-native RAN architectures also rely on multi-source telemetry to expose radio, traffic, and service context to intelligent controllers and AI/ML applications [24]. However, most existing telemetry frameworks are designed for specific domains, vendor interfaces, or monitoring tasks. Cross-domain data collection, time alignment, format consistency, repository management, and service-level interpretation across optical transport, fibre sensing, and wireless access domains remain insufficiently addressed.

B. Semantic Data Abstraction for Privacy-Aware Cross-Domain Fusion

Cross-domain dataset generation requires not only collecting heterogeneous telemetry, but also deciding which information should be shared, stored, and fused. Directly aggregating raw wireless telemetry into a central repository may introduce unnecessary communication and storage overhead while exposing device-specific or service-sensitive information. This issue is particularly critical in multi-access wireless networks, where raw telemetry may reveal device identity, access behaviour, traffic patterns, mobility states, and service conditions [25]. Therefore, practical optical–wireless data fusion requires compact and task-relevant representations that preserve useful information for downstream analysis while reducing the exposure of sensitive raw observations.

Task-oriented semantic communication provides a useful principle for such abstraction by transmitting information relevant to downstream tasks rather than reconstructing all source observations with equal fidelity [26, 27]. Within telemetry systems, semantic representations can be constructed from expert-designed indicators or learned encoders. In particular, information-bottleneck methods can learn compact latent variables that preserve task-relevant information while discarding irrelevant input details [28]. Such representations can summarise wireless load, link stability, access diversity, congestion risk, and service degradation, thereby reducing data volume and enabling more efficient fusion with optical transmission and fibre-sensing measurements. However, compact representations are not automatically privacy preserving, because sensitive attributes may still be inferred from the transmitted features. Privacy-aware representation learning therefore requires information bottlenecks, noise injection, adversarial objectives, or representation regularisation to balance downstream utility against sensitive information leakage [28–31].

Existing studies on semantic-communication and privacy-aware representation mainly focus on single-domain communi-

cation data, end-to-end transmission efficiency, or downstream task accuracy. Limited work has investigated how wireless telemetry should be abstracted for cross-domain optical–wireless dataset generation, or how such representations should be evaluated jointly in terms of cross-domain utility, data reduction, and privacy exposure. This motivates the design of wireless semantic modules that transform raw multi-access telemetry into compact task-relevant representations for privacy-aware fusion with optical transmission and fibre-sensing data.

C. Digital Twins and LLM-Driven Data Generation

The preceding subsections concern the collection and selective sharing of practical telemetry. Synthetic data address a different limitation: even a well-instrumented testbed cannot economically cover every topology, loading condition, device setting, or rare failure. Synthetic data can therefore complement, rather than replace, independently collected measurements by expanding scenario coverage and providing labels under controllable conditions. Network DTs provide a natural execution tool because they emulate network behaviour under configurable topology, traffic, device, channel, and service conditions [32].

In optical networks, GNPpy provides an open and vendor-independent physical-layer modelling framework for emulating optical paths under configurable topology, equipment, channel loading, and transmission-parameter settings [33]. Recent studies have used GNPpy-based DTs to generate optical-network datasets under normal and degraded conditions, such as amplifier degradation, span-loss variation, and component misconfiguration [34, 35]. In wireless networks, simulators and DT platforms, such as ns-3 and Colosseum, support repeatable packet-level, radio frequency, and protocol-stack emulation [36, 37], while generative-AI-enabled wireless DTs have been explored for 6G modelling and scenario augmentation [38, 39]. These studies demonstrate the value of DTs for controllable data generation. Nevertheless, DT-based dataset generation still requires substantial manual effort to define objectives, configure scenarios, execute simulations, validate results, and refine parameter settings. This manual process becomes costly and difficult to scale in cross-domain optical–wireless environments, where different DT tools rely on different abstractions, interfaces, and configuration formats.

Recent studies have begun to integrate LLMs, AI agents, and DTs to reduce the manual burden of network automation. In optical-network DT research, modelling, telemetry, self-learning, and lifecycle management have been identified as key components for intelligent optical networks [40]. GPT-enabled DT assistants have also been demonstrated to interpret user intentions and interact with DT functions for autonomous optical-network management [41]. In addition, DTs have been used to provide updated network states and pre-verify LLM-generated strategies before deployment, with demonstrations covering performance optimisation, protection switching, and recovery after fibre cuts [42]. Other studies have explored fine-tuned local agents, DT-aided wavelength-loading control, AI-agent integration, and field-trial demonstrations for optical-network lifecycle automation [43–46]. These studies demonstrate the potential of LLMs and AI agents to interact with DTs, but their focus remains on network operation and management rather than autonomous dataset generation.

The cited DT assistants and AI-agent studies automate parts of modelling and network operation, but they still rely on explicitly supplied objectives and configuration ranges [41–44]. This leaves dataset generation as a distinct workflow-design problem.

The present work therefore studies an LLM-driven interface that translates a high-level request into structured tasks, invokes optical and wireless DT engines, analyses generated KPIs, and proposes refinements. Section 5 evaluates this as a feasibility demonstration and includes a deterministic configuration-enumeration baseline; it does not claim that an LLM is necessary for exhaustive enumeration.

3. UNIFIED FRAMEWORK FOR CROSS-DOMAIN DATASET COLLECTION AND LLM-DRIVEN DATA GENERATION

Figure 1 illustrates a unified framework for cross-domain dataset collection and LLM-driven data generation in wireless–optical network environments based on our previous work [5] [6]. The bottom layer represents the physical infrastructure of a general end-to-end network scenario, in which wireless access domains are interconnected through an optical transport network. This abstraction captures a practical multi-domain networking environment where heterogeneous devices, protocols, and control entities coexist across wireless and optical segments.

The framework has two complementary branches. The practical-data branch collects and publishes independently acquired telemetry, while the generated-data branch converts a user request into simulator inputs and retained artefacts. Component-level storage and telemetry details are deferred to Section 4.

In this paper, the implemented path comprises telemetry export, semantic evaluation, orchestrated task construction, and GNPpy/ns-3 execution. AI engines and network controllers remain contextual integration points; QoT prediction, path optimisation, and physical-network enforcement by those blocks are not evaluated.

Figure 1 is therefore a scope map, whereas Fig. 6 later gives the evaluated data-generation sequence. The orchestrator dispatches that sequence but is not claimed to control every component in Fig. 1.

The detailed design of the cross-domain telemetry architecture across optical and wireless networks is presented in Section 4, while the LLM-driven multi-agent DT framework for cross-domain data generation is further elaborated in Section 5.

4. CROSS-DOMAIN TELEMETRY ARCHITECTURE ACROSS OPTICAL AND WIRELESS NETWORKS

A. Proposed Cross-Domain Telemetry Architecture

Figure 2 presents the cross-domain telemetry architecture developed from our earlier telemetry platform [5]. In this study, the implemented path comprises device-data acquisition, time-series and configuration storage, Kafka transfer, exported-record semantic evaluation, and the DT interfaces demonstrated in Section 5. The AI-engine, controller, local-inference, and closed-loop control blocks are prospective integration points. No scalability benchmark, controller deployment, or cryptographic privacy guarantee is claimed. This scope applies throughout this subsection.

A.1. Optical Transport Network Physical Layer

At the bottom of Fig. 2, the physical layer of the optical transport network consists of multiple interconnected optical nodes, including aggregation and intermediate nodes. These nodes represent practical optical transport infrastructure, in which

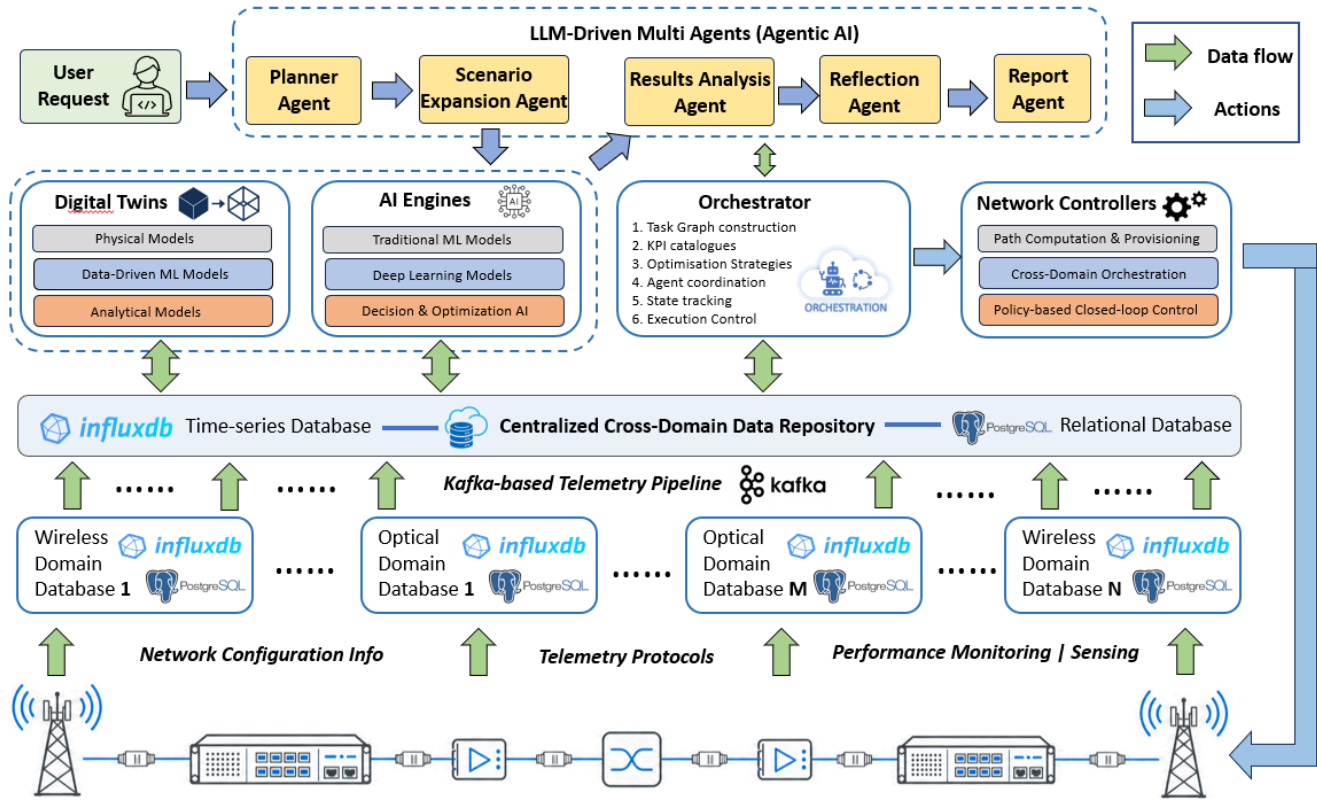


Fig. 1. Scope map of the complete dataset framework. The left branch organises independently acquired practical datasets; the right branch places the demonstrated DT data-generation workflow within a broader prospective network-intelligence context.

NEs such as transponders, fibre links, optical switches, reconfigurable optical add-drop multiplexers (ROADMs), and optical amplifiers generate heterogeneous operational data.

The collected optical domain data can be categorised into three types: performance monitoring metrics, equipment parameters and topology information, and fibre sensing data. Performance monitoring metrics include receiver-side BER, OSNR, received optical power, amplifier input/output power levels, and per-port optical power measurements from optical switches. Equipment parameters and topology information include transmitter-side launch power, wavelength allocation, modulation format, amplifier gain, node configuration, and link information. Fibre sensing data include SOP, chromatic dispersion (CD), polarisation mode dispersion (PMD), and backscatter traces. These data provide communication- and physical-layer visibility, supporting applications such as QoT estimation, anomaly detection, environment-aware operation, and proactive maintenance.

A.2. Distributed Node-Level Databases

Local databases store node- or domain-specific telemetry. Continuously sampled BER, OSNR, optical-power, and sensing streams are stored in InfluxDB, whereas topology and equipment configurations are stored in PostgreSQL.

A.3. Telemetry Service Design

The implemented telemetry service ingests available NETCONF, gRPC/gNMI, REST/HTTP, and vendor-interface records through Telegraf or source-specific collectors and writes timestamped values to the local databases.

For repository transfer, selected local records are published to Kafka topics and consumed by the central repository. Campaign

and source identifiers are retained during aggregation so that independently acquired records are not mistaken for naturally aligned observations.

Before wireless data are shared with the central repository, the evaluated semantic modules transform raw telemetry into compact task-related representations. Their design and evaluation are detailed in Section B.

A.4. Centralised Cross-Domain Data Repository

The central repository provides a common access point for selected monitoring streams, topology, equipment, and service metadata. Importantly, a common repository does not imply that all released campaigns are time aligned: timestamps and campaign identifiers are retained so users can distinguish contemporaneous records from independently acquired datasets.

A.5. Network Intelligence and Controllers

Above the repository, DTs provide the execution environments used in Section 5; AI engines and network controllers show potential consumers of practical and generated data.

A.6. Wireless Access Network Domains

The wireless-domain blocks in Fig. 2 show where 5G, Wi-Fi, and LiFi testbed telemetry enters the common repository and semantic interface.

B. Semantic Modules for Cross-Domain Data Fusion

The semantic module is evaluated as an interface for converting wireless telemetry into compact representations. We compare V1, a hand-crafted representation; V2, a learned eight-dimensional latent representation; and V3, a learned three-dimensional bottleneck with optional training-time noise. The term privacy-aware

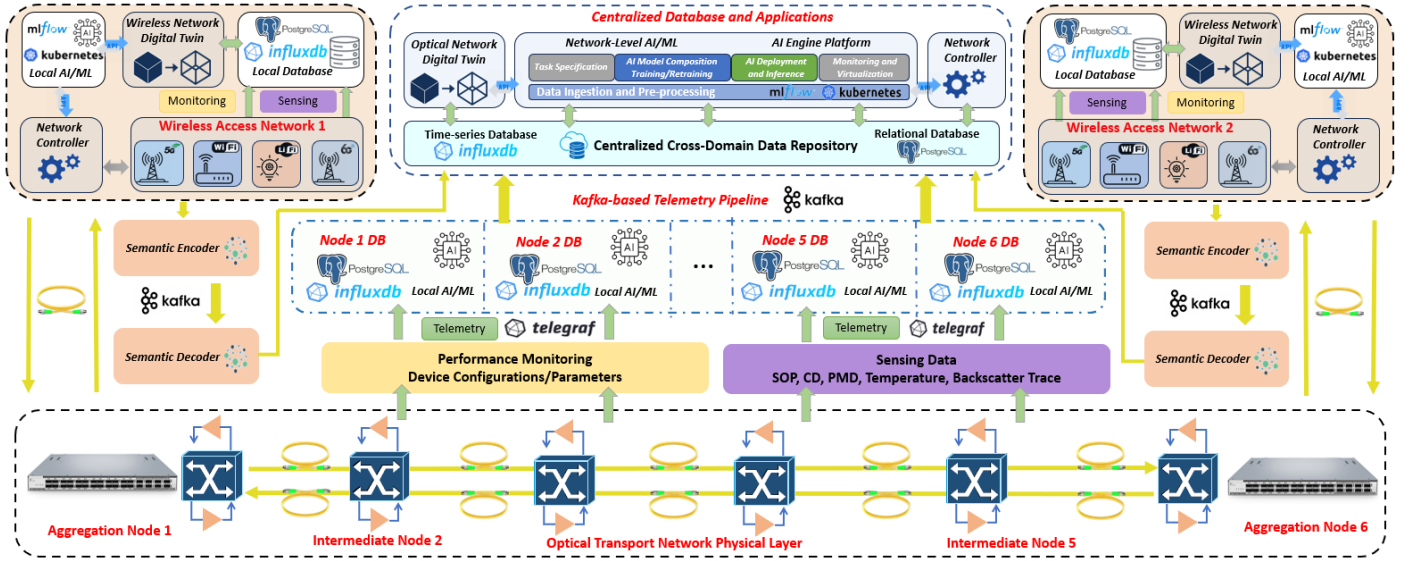


Fig. 2. Telemetry and repository view of the framework. Solid data paths correspond to implemented collection/storage functions; DT, AI-engine, and controller blocks indicate integration points, not evaluated closed-loop functions in this paper.

refers to an evaluated design objective—dimensionality and noise are varied and attribute leakage is measured—rather than a formal privacy guarantee.

B.1. Semantic Module Designs

The input contains 12 features derived from 30-s wireless records: aggregate 5G, Wi-Fi, and LiFi traffic; total traffic; mean Wi-Fi and LiFi signal levels; mean Wi-Fi inactivity; a five-record rolling traffic fluctuation; three RAT-activity indicators; and a combined signal-quality score. RAT activity is defined relative to the 25th percentile of training traffic. All min-max and standard scalars are fitted on the training period only. Four task-oriented targets are then formed: load L , access diversity D , link stability S , and congestion risk C . With q denoting combined signal quality, f normalised traffic fluctuation, and i normalised inactivity,

$$D = (a_{5G} + a_{WiFi} + a_{LiFi})/3, \quad (1)$$

$$S = [0.5q + 0.3(1 - f) + 0.2(1 - i)]_{[0,1]}, \quad (2)$$

$$C = \sigma(2L + 1.5i + f - 1.5S - 0.8D), \quad (3)$$

where $[\cdot]_{[0,1]}$ denotes clipping and $\sigma(\cdot)$ is the logistic function. These variables were selected to cover traffic intensity, temporal stability, multi-access availability, and a composite risk indicator using fields present across the wireless testbed exports. A confidence value $1 - f$ accompanies each transmitted representation. Fig. 3 summarises the three designs.

V1 transmits eight deterministic indicators: L , S , D , C , rolling-mean traffic, traffic fluctuation, signal quality, and inactivity risk. V2 uses a 12–64–32–8 encoder and a symmetric decoder to reconstruct L , S , D , and C . V3 uses a 12–32–16–3 encoder and a symmetric decoder. Its primary setting injects zero-mean Gaussian noise with $\sigma = 0.05$ at the encoder input during training. The encoder-decoders use mini-batch Adam, batch size 32, at most 200 epochs, and early stopping with patience 12 on validation loss. For downstream utility, the four decoded targets plus confidence are fused with optical features; the privacy attacker receives the latent code plus confidence. The source, exact seeds, preprocessing parameters, and per-seed outputs are released with the experiment scripts.

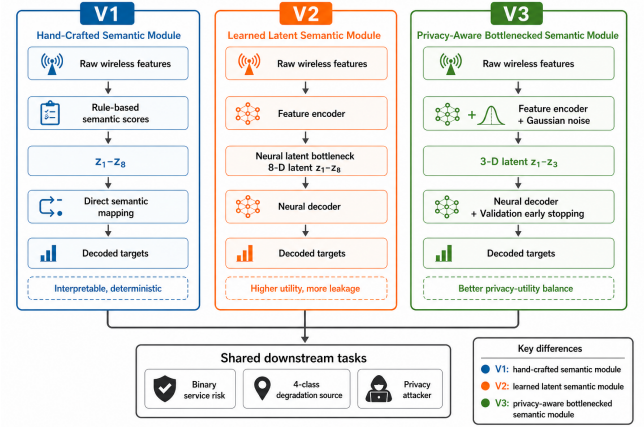


Fig. 3. Structures of three semantic-module designs.

The optical and wireless campaigns do not overlap in time. We therefore construct an explicitly labelled interoperability benchmark by mapping the ordered 20,960 optical records to the ordered 2,849 wireless records at uniformly spaced relative positions; no timestamp equality is asserted. This benchmark tests schema compatibility and representation utility, not contemporaneous end-to-end behaviour. After one-step target shifting, 2,848 samples remain. Optical risk is defined using the training-period 15th percentile of Q-factor or 85th percentile of BER. Wireless risk is defined as $C > 0.65$, $S < 0.35$, or $D < 0.34$. Binary service risk is their logical OR; four degradation classes denote neither, wireless only, optical only, and both. These are rule-derived benchmark labels rather than field-observed service failures.

B.2. Testing Results

Samples are partitioned chronologically into 60% training, 20% validation, and 20% test segments, with a ten-record purge gap before each boundary to prevent five-record rolling features crossing partitions. This yields 1,698/560/570 evaluated samples. Every threshold, scaler, encoder, and downstream classifier

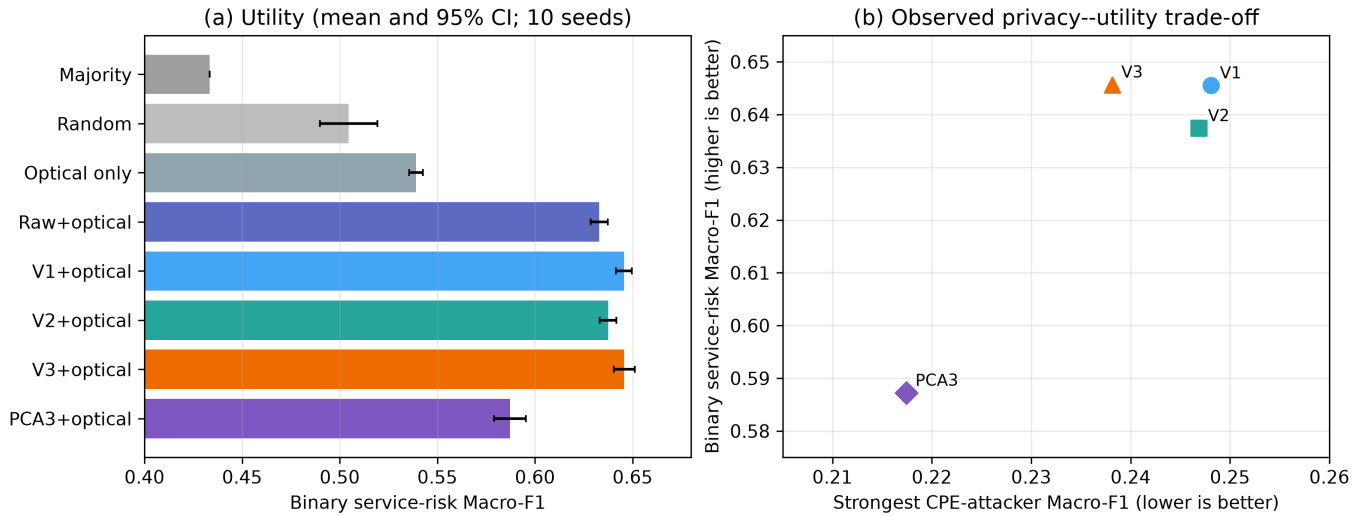


Fig. 4. Leakage-free temporal evaluation. (a) Binary service-risk utility; error bars are 95% confidence intervals over ten seeds. (b) Utility versus the strongest observed CPE attacker among four attacker families. Lower attacker Macro-F1 indicates less observed leakage but is not a formal privacy guarantee.

Table 1. Temporal-holdout utility and CPE-attribute leakage (mean $\pm$ 95% confidence interval over ten seeds).

Representation	Binary service-risk Macro-F1	Four-class Macro-F1	Strongest CPE-attack Macro-F1↓
Majority baseline	0.4334 $\pm$ 0	0.2167 $\pm$ 0	0.1235 $\pm$ 0
Stratified-random baseline	0.5046 $\pm$ 0.0147	0.2427 $\pm$ 0.0101	0.2530 $\pm$ 0.0095
Raw wireless + optical	0.6329 $\pm$ 0.0044	0.4161 $\pm$ 0.0051	0.2757 $\pm$ 0
V1 + optical	0.6455 $\pm$ 0.0040	0.3823 $\pm$ 0.0033	0.2481 $\pm$ 0
V2 + optical	0.6374 $\pm$ 0.0044	0.3958 $\pm$ 0.0175	0.2469 $\pm$ 0.0102
V3 + optical ($\sigma = 0.05$)	0.6457 $\pm$ 0.0054	0.3955 $\pm$ 0.0185	0.2382 $\pm$ 0.0113
PCA3 + optical	0.5872 $\pm$ 0.0081	0.3829 $\pm$ 0.0045	0.2174 $\pm$ 0.0059

is fitted after the split using training data only. Utility is measured with a 300-tree class-balanced random forest over ten seeds (3, 7, 11, 19, 23, 31, 42, 53, 71, and 97). We report Accuracy, Balanced Accuracy, Macro-F1, AUROC, and AUPRC; Table 1 shows the primary Macro-F1 results and 95% confidence intervals.

Privacy is operationalised as attribute inference from the transmitted representation. Dominant CPE and dominant RAT are defined by the largest aggregate traffic contribution in each record. Four attackers are evaluated: class-balanced logistic regression, a 300-tree random forest, a two-layer MLP (32 and 16 units), and histogram gradient boosting. Raw telemetry, PCA, random projection, majority, and stratified-random baselines are included. Because the test set contains 559 Wi-Fi and only 11 LiFi records (and no 5G record), RAT accuracy is dominated by the 0.981 majority baseline and is retained only as a diagnostic. CPE Macro-F1 is the primary privacy metric.

V3 reaches binary Macro-F1 0.6457 ± 0.0054 , similar to V1 (0.6455 ± 0.0040) and above raw wireless plus optical features (0.6329 ± 0.0044). The overlapping intervals do not support a claim that V3 is superior to V1. For the four-class task, raw features perform best. Moreover, the test distribution is 436/115/13/6, so the four-class result is exploratory. Removing V3 training noise slightly increases binary Macro-F1 from 0.6457 to 0.6473; increasing noise to $\sigma = 0.10$ lowers the strongest CPE-attack Macro-F1 to 0.2319. Thus, noise reduces observed leakage but does not improve utility in this dataset.

V3 therefore offers a useful operating point: it retains V1-level binary utility with moderately lower observed CPE leakage. It is

not universally optimal—PCA3 leaks less but loses more utility, and raw features perform best on the imbalanced four-class task. We consequently describe V3 as privacy-aware and report the complete trade-off rather than claiming privacy preservation or statistical superiority.

C. Practical Dataset Description

Under the telemetry architecture, we openly provide datasets from separate optical-transport, fibre-sensing, and wireless-access campaigns [47]. Table 2 summarises their scope and machine-readable files. The campaigns have different dates, sampling rates, and service contexts; “cross-domain” therefore refers to common publication and representation rather than native sample-level alignment. The datasets and associated meta-data released in the practical-data and revision repositories are licensed under Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0), which permits sharing and adaptation with attribution for non-commercial purposes. We therefore describe these resources as openly available. Software and identified third-party material retain their own terms. A Croissant 1.0-compatible JSON-LD record documents the released data, access locations, sizes, checksums, and source-defined units. The final synthetic wireless bundle is available under `W4C_THREE_RAT_SWEEP` / [48].

C.1. One-Month Performance Monitoring Data from Optical Transport Network over NDFF

Figure 5(a) presents the experimental setup over part of the UK National Dark Fibre Facility (NDFF), including transponders,

Table 2. Summary of the practical, constructed, and synthetic evaluation datasets used in this paper. Sizes refer to the principal processed files used in the revision experiments; raw sensing traces are distributed as multiple files.

Dataset	Records/sampling	Principal fields or labels	Format/processed size	Access/reuse terms
339-km NDFF monitoring (2023–24)	1,373 processed records from 26 Dec. 2023 to 26 Jan. 2024; 10-s raw acquisition resampled to 15 min	Four Voyager BER channels and four derived Q-factors with timestamps	CSV and scripts; 0.15 MB	CC BY-NC 4.0
339-km NDFF Voyager (2025–26)	20,960 valid raw records from 22 Dec. 2025 to 24 Jan. 2026; mostly 70–72-s spacing with acquisition gaps	Eight channel BER ratios and real timestamps; derived Q-factor used in semantic experiments	Raw CSV; 1.79 MB	CC BY-NC 4.0
986-km NDFF QoT	Two links and ROADM nodes; record count and time coverage not available in the current release	12 channel states and Q-factors at three nodes; units follow the cited source	External CSV source cited in YangOFC2023; file size not available here	External reference; not in current release
Urban fibre sensing	Four-week campaign over 7.776-km and 8-km links; exact record count not available in the current release; polarimeter sampling at 97.656 kSa/s	BER, Q-factor, OSNR, received power, S_1 – S_3 , oscilloscope time/voltage; field units follow the source	MAT and CSV; file sizes not available here	External reference; not in current release
Wireless access	2,849 records over 24 h (nominal 30-s telemetry); 46 columns	CPE traffic, 5G temperature/traffic/power, Wi-Fi rate/inactivity/signal, LiFi rate/signal; raw units not declared in source	CSV; merged CSV 0.79 MB	CC BY-NC 4.0
Synthetic GNP optical DT	8,192 executions; 6,144 unique configurations (6,120 valid and 24 no-signal); 96-case replay	Four paths, two modulation/spectrum settings, 256 binary occupancy patterns, three launch powers; GSNR, status, warnings, and replay differences	CSV, JSON, and JSONL; principal 6,144-row table 10.09 MB	CC BY-NC 4.0
Synthetic three-RAT wireless DT	48 configurations; 432 application-flow records (144 each for NR, Wi-Fi 802.11ax, and LiFi); 10-s runs	Per-flow KPIs (Mbps, ms, packets, bytes, ratios) and LiFi optical-state variables (m, rad, W, A, ratios); loads 2/5/10/20 Mbps, speeds 0.5/3 m/s, FOV 40°/70°, seeds 1–3	ns-3.44/5G-LENA; CSV, XML, JSON; public three-RAT bundle	CC BY-NC 4.0
Constructed fusion benchmark	2,848 one-step prediction samples; chronological split with 20 purged boundary records	Current/target source-row indices, deterministic wireless semantics, optical statistics, training-only thresholds, and rule-derived labels	CSV; 1.60 MB	CC BY-NC 4.0

The final synthetic wireless fields and units are listed in the public synthetic-wireless data dictionary. For source files not present in the current repository, record counts, file sizes, and field units are not inferred from citations or filenames.

erbium-doped fibre amplifiers, wavelength-selective switches, and optical switches. Seven equalised 200-GHz-spaced PM-16QAM/QPSK signals were generated by Facebook Voyager, ADVA FSP 3000 TeraFlex, and Ciena transponders. Voyager and ADVA support 16QAM and QPSK, whereas the Ciena device supports QPSK. The signals traversed the 339-km loop from Bristol through Bradley Stoke, Froxfield, Reading, and Power Gate, with carrier frequencies from 193.25 to 194.45 THz. The launch power was fixed at -5 dBm as the operating point used for this campaign; no optimality claim is made. A cloud configuration and monitoring database stored the field-trial measurements.

The public repository distinguishes two measurement campaigns on this same 339-km loop. The earlier release contains 1,373 processed records collected from 26th December 2023 to 26th January 2024. Its 10-s raw BER measurements were resampled to 15-minute intervals, and four Voyager-channel Q-factors were calculated using $Q = \sqrt{2} \operatorname{erfcinv}(2 \operatorname{BER})$. This preprocessing reduces short-term measurement noise and aligns the sampling interval with that of the Microsoft open-source dataset.

A separate Christmas 2025 campaign, used as the optical source for the constructed semantic-fusion benchmark in Section IV-B, contains 20,960 valid eight-channel BER records from 22nd December 2025 16:38:43 to 24th January 2026 16:10:55. These are real acquisition timestamps retained without shifting or anonymisation. The raw CSV preserves 38 embedded headers from 39 concatenated acquisition segments; the released preprocessing removes those header rows before timestamp and BER parsing. Consecutive valid records are mostly 70–72 s apart, with larger acquisition gaps, and this file was not resampled to 15-minute intervals. Its unchanged source file and SHA-256 checksum are published with the acquisition note [47]. Because the two campaigns differ in date, channel count, and sampling, they are reported as separate products rather than as stages of one transformation.

We also release a complementary dataset from a previous 986-km NDFF field trial [49] comprising two fibre links and

ROADM nodes, where multi-channel Q-factor measurements were collected for end-to-end QoT prediction. Together, the 339-km and 986-km datasets provide representative short- and long-link field-trial scenarios for QoT estimation, short-term performance forecasting and anomaly detection.

C.2. Four-Week Sensing Data from BDFI and M Shed

Figure 5 (b) illustrates the experimental setup for long-term fibre sensing based on bidirectional transmission. The experiment was conducted over two deployed urban fibre links in Bristol: a 7.776-km link connecting our lab in Merchant Ventures Building (MVB) to Bristol Digital Futures Institute (BDFI), and an 8-km link connecting MVB to M Shed Museum. Along the MVB-BDFI route, the fibre passes beneath multiple railway tracks near Bristol Temple Meads Station at the 3.2-km point, where it laterally traverses beneath a railway bridge. In contrast, the MVB-M Shed route does not pass beneath railway tracks.

An ADVA FSP 3000 Teraflex transponder generated two 32-Gbaud PM-QPSK optical signals, which were launched into the fibre links through optical circulators to enable counter-propagating transmission. The endpoint Circulators were interconnected through the live field fibre, with the forward and backward channels centred at 193.4 THz and 191.3 THz respectively. At each endpoint, the received optical signal was split using an optical coupler: 90% of the optical power was directed to the coherent receiver, while the remaining 10% passed through a polariser and was delivered to a polarimeter and a real-time oscilloscope equipped with a photodetector. This configuration enables simultaneous communication monitoring and fibre sensing. The polariser projects the received state of polarisation (RSOP) onto a fixed eigenstate, allowing environmental disturbances along the fibre to be captured through the measured polarisation dynamics. For vibration localisation, a common trigger from an arbitrary waveform generator was used to synchronise both the polarimeter and oscilloscope.

The dataset spans four weeks and is provided in MAT and CSV formats. It contains communication KPIs, including BER,

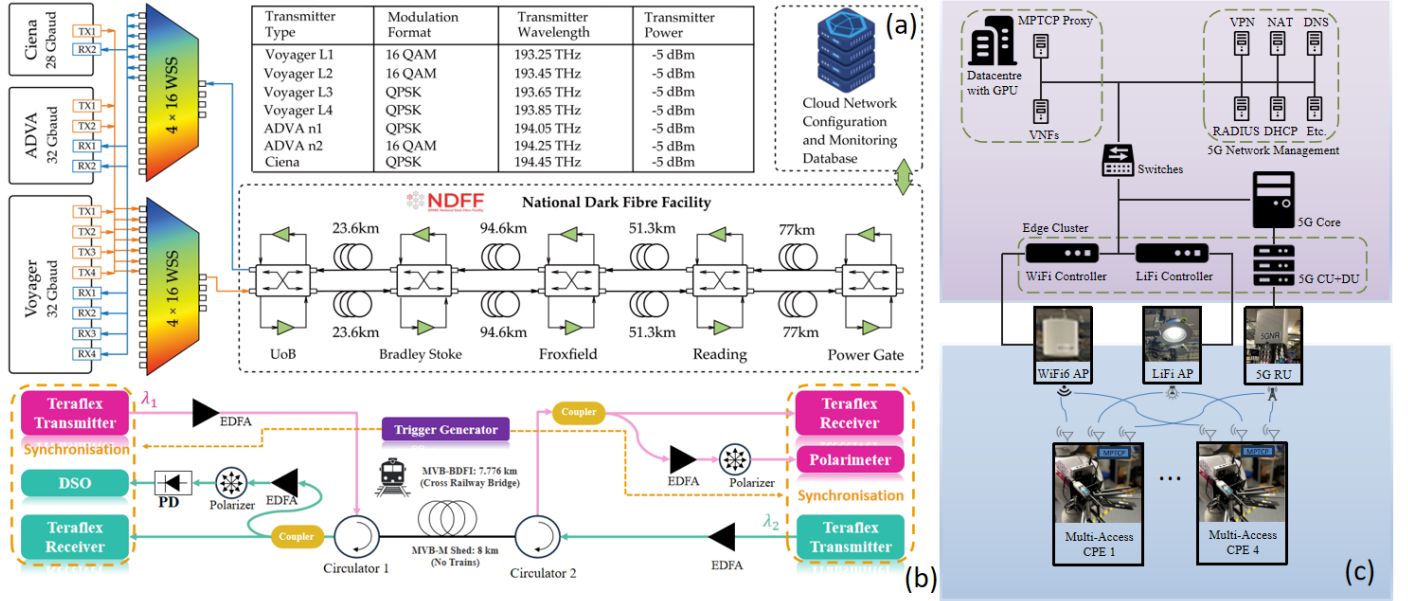


Fig. 5. Experimental setups for practical dataset collection: (a) optical transmission performance monitoring over the NDF, (b) bidirectional fibre sensing over deployed urban fibre links, and (c) multi-access wireless network testbed.

Q-factor, and OSNR, and polarisation telemetry (Stokes parameters S_1 , S_2 , and S_3). These measurements were continuously collected by the ISIC end-to-end monitoring platform. Because the data originate from deployed urban fibres rather than laboratory spools, they capture environmental and operational variation. The dataset supports studies of SOP evolution, vibration detection and localisation, communication degradation under physical disturbance, ML-based fibre sensing, and communication-sensing integration.

C.3. 24-Hour Status and Performance Data from a Wireless Access Network Testbed

The wireless dataset was collected from a wireless access network testbed originally developed in the 5G-CLARITY project and further extended in the REASON project. These radio signals and telemetry records originate entirely within that wireless testbed; they are not converted from, transported as, or synchronised to signals on either optical field link. The optical interconnection shown in Figs. 1 and 2 is a system-level architecture, not the signal source for this campaign. As shown in Fig. 5(c), the testbed integrates 5G New Radio, Wi-Fi 6, and LiFi. Four fixed CPEs were deployed: CPE1–CPE3 were dual-homed to 5G and Wi-Fi, while CPE4 was tri-homed to 5G, Wi-Fi, and LiFi. All CPEs used MPTCP v1 in fullmesh mode with the default BLEST scheduler. During collection, CPE1–CPE3 generated randomised iPerf traffic towards CPE4.

The dataset provides 24 hours of time-aligned wireless telemetry, covering per-interface CPE traffic, 5G cell metrics, LiFi cell metrics, and Wi-Fi cell metrics. Specifically, it includes per-interface traffic distribution, 5G temperature, throughput and power, LiFi transmit/receive rates and signal power, and Wi-Fi bitrates, inactivity time and average signal levels. This dataset offers a comprehensive view of heterogeneous wireless performance and supports research on AI-driven multi-access optimisation, multi-connectivity control, MPTCP scheduling, and cross-technology coordination in 5G and beyond networks. By capturing the complementary characteristics of 5G, Wi-Fi, and LiFi in a unified testbed, it also provides a basis for wireless

DTs, semantic telemetry abstraction, and cross-domain optimisation with optical datasets.

5. LLM-DRIVEN MULTI-AGENT DIGITAL TWINS FOR AUTOMATED CROSS-DOMAIN DATA GENERATION

Practical telemetry cannot cover every topology, loading condition, or rare event. This section presents an LLM-driven multi-agent DT workflow as a feasibility demonstration for generating complementary synthetic data.

A. Proposed LLM-Driven Multi-Agent Digital Twin Framework

Figure 6 shows the operational definition of “multi-agent” used here: five schema-defined functional roles coordinated through orchestrator-owned shared state. Planner, Scenario Expansion, and Reflection invoke an LLM; Results Analysis and Report are deterministic. Topology and equipment files, KPI definitions, simulator wrappers, and stored execution results provide the roles’ knowledge and tools. The orchestrator passes a natural-language request through planning, scenario expansion, DT execution, analysis, optional refinement, and reporting.

The reported workflow invokes GNPpy and ns-3 as DT engines and retains structured requests, configurations, results, and logs. It demonstrates that the role sequence can produce auditable simulator artefacts; it does not establish that every role requires an LLM or that the decomposition is more efficient than a conventional script.

B. Evaluation Protocol and Reproducibility Scope

We compare the GNPpy demonstration with an independent deterministic enumerator over four destinations, two modulation/slot-width settings, 256 occupancy patterns, and three launch powers. Configuration keys exclude iteration and scenario identifiers, allowing repeated executions to be separated from the 6,144 unique cases. The script enumerates the Cartesian product only; it does not perform intent parsing or optical propagation.

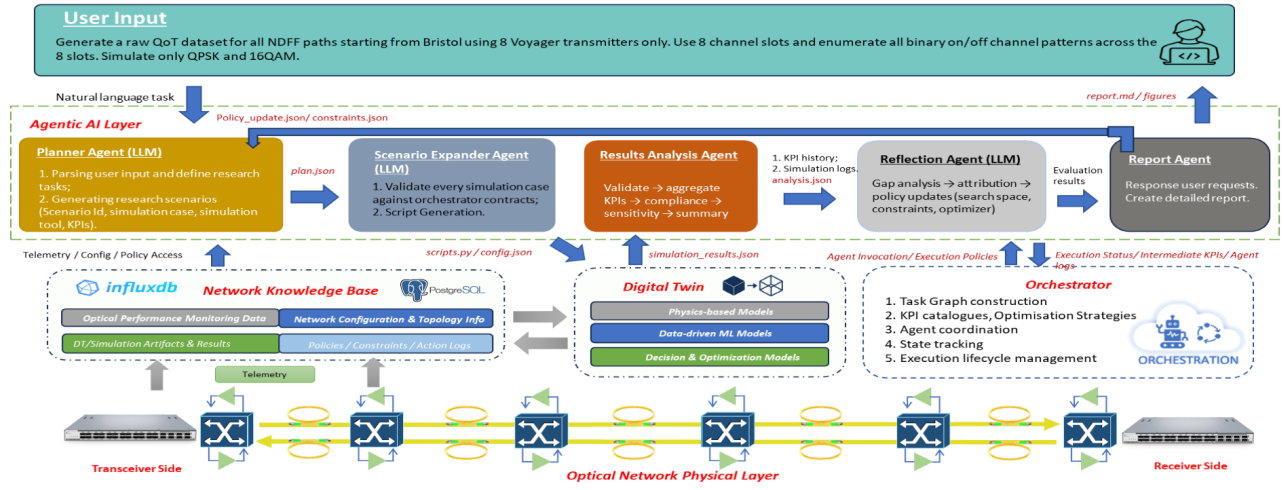


Fig. 6. Evaluated data-generation sequence. Fig. 1 provides the framework scope, whereas this figure shows the shared task state, role-specific LLM invocations, simulator interface, and retained artefacts.

Table 3. Instrumented OpenAI feasibility run.

Item	Observed value
Model/settings	gpt-4o-mini-2024-07-18; temperature 0.2; time-out 90 s; maximum three attempts; strict mode
Role calls	Planner 2; Scenario Expansion 2; Reflection 2
API outcomes	6/6 successful first attempts; 0 retries; 0 API failures
Token usage	106,041 prompt; 2,267 completion; 108,308 total
API latency	26.22 s summed; 2.88–7.61 s per call
Estimated cost	\$0.01727 using the execution-date pricing snapshot
Retained evidence	Rendered requests/responses, call log, recovery record, and source/input hashes

Table 4. Configuration-set baseline for the corrected GNPY demonstration.

Method	Executions	Unique	Duplicate rate
Deterministic enumerator	0	6,144	0
LLM-orchestrated history	8,192	6,144	25%

Both sets have precision and recall 1.0 against the independently specified 6,144-key set. The deterministic baseline performs configuration enumeration only, uses no LLM/API calls, and is not an end-to-end runtime comparison.

For reproducibility, a separately labelled instrumented run used gpt-4o-mini-2024-07-18, temperature 0.2, a 90-s timeout, at most three attempts per call, and strict mode. Planner, Scenario Expansion, and Reflection were each called twice. All six API calls succeeded on their first attempt, using 108,308 tokens, 26.22 s summed API latency, and an estimated \$0.01727 in text-token cost. Complete prompts, responses, logs, recovery history, and source/input hashes are released at commit 82914127b70acf3b220bc09a56a179cb7abd1252 [50]. Manual recovery was required during simulator execution; accordingly, these measurements describe one feasibility run rather than unattended reliability or agent superiority.

C. Use Cases and Demonstration Results

We demonstrate the workflow with GNPY for optical transport and ns-3 for wireless access. The final synthetic wireless source and results are released under W4C_THREE_RAT_SWEEP/ [48]; the upstream NS3_Wireless/ directory is an earlier baseline.

C.1. GNPY-Based Network-Level QoT Data Generation

In this use case, the workflow generates an optical QoT artefact based on the NDDF topology. The request covers paths from Bristol, Voyager transmitters, eight WDM slots, and QPSK and 16QAM configurations. The Planner forms the study specification, and Scenario Expansion constructs the candidate paths, binary channel-occupancy patterns, and launch-power settings for GNPY execution.

As illustrated in Fig. 7(a), four paths, two joint modulation/spectrum configurations, and $2^8 = 256$ occupancy patterns produce 2,048 cases per launch power. The first pass contains 2,048 records at -5.5 dBm; the refined pass contains 6,144 records at -6.5 , -6.0 , and -5.5 dBm. The combined 8,192 execution records reduce to 6,144 unique configurations because the first-pass keys recur in the refined pass. Of the unique configurations, 6,120 return numerical results and 24 are expected all-off, no-signal cases. The observed set has no missing or additional keys relative to the deterministic enumerator.

Figure 7(b)–(d) uses the unique refined-pass records. At -5.5 dBm, mean QPSK GSNR decreases from 26.34 to 22.57 dB on Bristol–Bradley Stoke and from 9.52 to 3.80 dB on Bristol–Powergate as active-channel count rises from one to eight. Launch-power sensitivity is route dependent: over the tested range, the Bradley Stoke mean decreases whereas the Powergate mean increases slightly. These three operating points do not establish a globally optimal launch power.

For numerical reproducibility, we independently replayed 96 cases stratified by path, joint modulation/slot-width setting, power, and channel load. A pinned GNPY v2.12 compatibility environment completed all 96 cases and reproduced the archived GSNR values within 0.00050 dB. A schema-corrected GNPY v2.14.2 environment also completed all cases but differed from the archive by 0.628 dB on average and 1.435 dB at most. The version sensitivity is larger on multi-span routes. Configuration warnings persist, so successful execution establishes numeri-

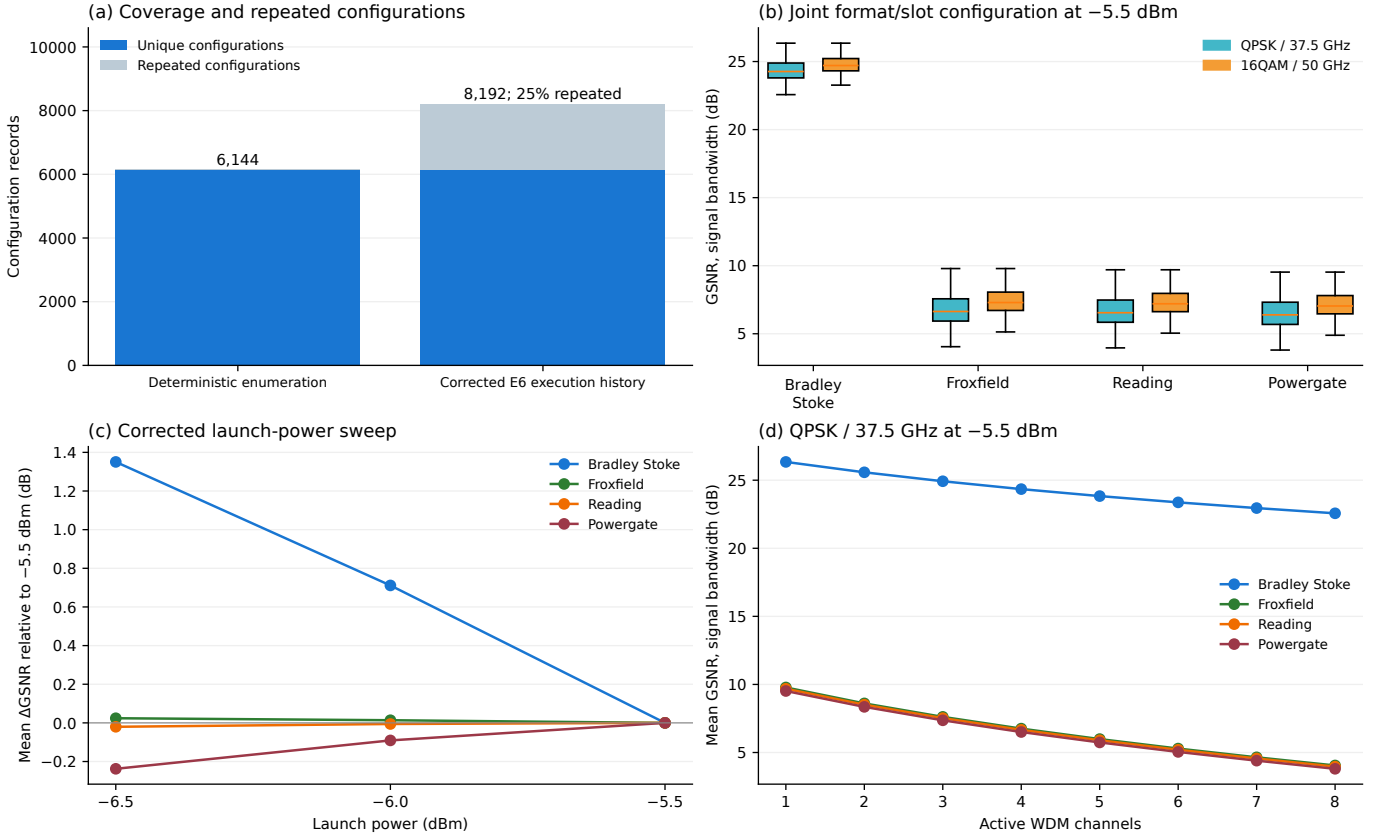


Fig. 7. GNPpy feasibility demonstration over the NDFF topology. (a) Configuration enumeration and execution-history counts; (b) GSNR distributions for the two modulation/slot-width settings at -5.5 dBm; (c) route-dependent mean GSNR change across the launch-power sweep; and (d) QPSK mean GSNR versus active-channel count. Numerical panels use 6,120 valid unique configurations; 24 all-off cases are excluded from metric summaries. GSNR is measured over the signal bandwidth.

Table 5. GNPpy replay over 96 stratified cases.

Environment	Success	Mean $ \Delta\text{GSNR} $	Max $ \Delta\text{GSNR} $
v2.12 compatibility	96/96	0.00015 dB	0.00050 dB
v2.14.2 schema-fixed	96/96	0.628 dB	1.435 dB

cal reproducibility under the stated environments rather than warning-free physical calibration.

QPSK and 16QAM use 37.5- and 50-GHz slots, respectively; their observed differences therefore reflect a joint modulation/spectral-spacing setting rather than a modulation-only effect.

Overall, the use case demonstrates conversion of a high-level request into structured simulator artefacts and a reflection-proposed refined sweep. The deterministic baseline shows that an LLM is unnecessary for exhaustive enumeration of this fixed space. Larger topologies, repeated requests, and calibrated equipment models remain future work.

C.2. ns-3-Based Multi-Access Network Performance Data Generation

The LLM-driven workflow constructs and executes a bounded three-access synthetic wireless scenario at the same technology-class level as the practical multi-access testbed: 5G NR, Wi-Fi 6/IEEE 802.11ax, and LiFi. The generated dataset is separate from the measured wireless telemetry. It demonstrates configurable synthetic data generation and workflow interoperability, not calibrated emulation of the physical testbed and not valida-

tion of measured wireless values.

Each scenario contains three common mobile CPEs, each carrying one interface for each access branch. The NR branch uses CTTC 5G-LENA version 4.0.y on ns-3.44 with one gNB and three UEs, a 3.5-GHz carrier, 40-MHz bandwidth, numerology 1, 30-dBm gNB transmit power, and the 3GPP InH-OfficeOpen channel. The Wi-Fi branch uses native WIFI_STANDARD_80211ax with one AP and three stations, a 5-GHz 20-MHz channel, fixed 16-dBm transmit power, and an explicit constant-rate HeMcs0 configuration. CPE mobility is RandomWalk2d within 50×50 m bounds at 0.5 or 3 m/s. All branches use UDP downlink traffic with 1,024-byte packets, a common offered load of 2, 5, 10, or 20 Mbps per application flow, a 1.0–9.5 s application window, and a 10-s simulation stop.

The LiFi branch is an equation-driven LOS LiFi/OWC link model. It computes Lambertian emission, transmitter/receiver geometry, receiver FOV, received optical power, photodetector responsivity, shot and thermal noise, electrical SNR, OOK BER, and packet-error probability. A stock ns-3 point-to-point packet path is used as the IP/UDP access abstraction, while packet corruption is driven by the optical link calculation. This is not a complete standardised LiFi PHY/MAC and its explicit parameters are uncalibrated to the practical testbed.

For half-power semi-angle Φ_{half} , detector area A , distance d , irradiance angle ϕ , incidence angle ψ , filter gain T_s , and concentrator gain $g(\psi)$, the Lambertian order is $m =$

$-\ln(2)/\ln(\cos \Phi_{\text{half}})$ and

$$H_{\text{LOS}} = \frac{(m+1)A}{2\pi d^2} \cos^m(\phi) T_s g(\psi) \cos(\psi), \quad (4)$$

when the irradiance geometry is valid and ψ is within the receiver FOV; otherwise $H_{\text{LOS}} = 0$. The concentrator is $g(\psi) = n^2 / \sin^2(\Psi_c)$ within the FOV, and received optical power is $P_r = P_t H_{\text{LOS}}$. The receiver current and the shot/thermal-noise terms determine SNR, which is mapped to BER = $Q(\sqrt{\text{SNR}}/2)$ and PER = $1 - (1 - \text{BER})^{8L}$. These equations follow the standard LOS optical-wireless link-budget formulation [51, 52] and are implemented in the source archived under W4C_THREE_RAT_SWEEP/ns3/[48].

Four offered loads, two mobility speeds, two LiFi receiver FOVs (40° and 70°), and seeds 1, 2, and 3 yield $4 \times 2 \times 2 \times 3 = 48$ run-level configurations. The 40° setting was selected by a preflight over 30° , 40° , 50° , and 70° ; it produced measurable out-of-FOV states without near-total outage under the declared geometry. Each final configuration produces nine application flows (three branches $\times$ three CPEs), giving 432 application-flow records. The 624 raw FlowMonitor rows additionally include infrastructure/control flows, which are excluded from application statistics. Outputs comprise FlowMonitor XML, per-flow KPI CSV, scenario/configuration metadata, explicit flow mappings, sampled LiFi optical states, and aggregated CSV files.

KPI Definitions and Interpretation. For received bytes B_{rx} , transmitted/received packet counts N_{tx} and N_{rx} , and FlowMonitor time quantities in seconds, the exported throughput is $8B_{\text{rx}}/[10^6(t_{\text{lastRx}} - t_{\text{firstTx}})]$ Mbps. Mean delay is $10^3 \text{delaySum}/N_{\text{rx}}$ ms; mean jitter is $10^3 \text{jitterSum}/(N_{\text{rx}} - 1)$ ms for $N_{\text{rx}} > 1$. The FlowMonitor-reported packet-loss ratio is $\text{lostPackets}/N_{\text{tx}}$. Because traffic and simulation terminate at finite times, this field is not interpreted as terminal packet loss; 172 of 432 application rows have $N_{\text{tx}} - N_{\text{rx}} \neq \text{lostPackets}$.

Validation of the Synthetic Wireless Dataset. Successful execution alone does not establish generated-data correctness. We therefore evaluated the final stack at the backend, equation, integrated-flow, repeatability, and full-sweep levels (Table 6). The NR/5G-LENA and native 802.11ax branches passed build and runtime checks. The independent LiFi mathematical tests passed 9/9 checks, and compiled ns-3 link-probe values agreed with the independent Python reference in 15/15 comparisons across aligned-near, aligned-far, and out-of-FOV geometries. The integrated acceptance run produced three flows per branch and 9/9 mapped application flows.

The final-source configuration was then executed twice with the same ns-3.44 build, seed, run number, workload, mobility, and FOV. The two outputs were byte-identical for the run configuration, scenario metadata, flow mapping, KPI CSV, FlowMonitor XML, and optical-state CSV. Finally, the bounded sweep completed all 48 configurations, with 432/432 application-flow mappings passing and no missing, extra, or duplicate configuration keys. These checks support correctness and reproducibility of the generated synthetic telemetry under the declared assumptions; they do not establish physical-testbed calibration or physical-model fidelity.

Under the adopted simulation configuration, increasing the offered load from 2 to 20 Mbps increased run-level throughput by 18.486 Mbps for NR and 15.726 Mbps for LiFi. For Wi-Fi 802.11ax, the corresponding throughput increase was only 0.151 Mbps, while mean delay increased by 1,653.88 ms

Table 6. Validation of the ns-3 synthetic wireless dataset.

Validation	Coverage	Result
Backend qualification	5G-LENA NR and native 802.11ax branches	Build/runtime PASS in the qualified ns-3.44 environment
LiFi equation tests	Independent Python reference; 9 tests	9/9 PASS
Compiled LiFi reference	3 geometries; 15 values	15/15 comparisons PASS
Integrated three-RAT mapping	1 final-source acceptance; 9 application flows	3 NR + 3 Wi-Fi + 3 LiFi; 9/9 mapped
Same-seed repeatability	1 final-source configuration executed twice	Byte-identical KPI, XML, mapping, metadata, and optical-state outputs
Full final sweep	48 configurations; 624 raw / 432 application flows	48/48 complete; 432/432 mappings pass

The rows distinguish backend qualification, independent equation/reference checks, integrated flow mapping, fixed-seed repeatability, and bounded-sweep completion under the declared scenario. None is physical-model validation.

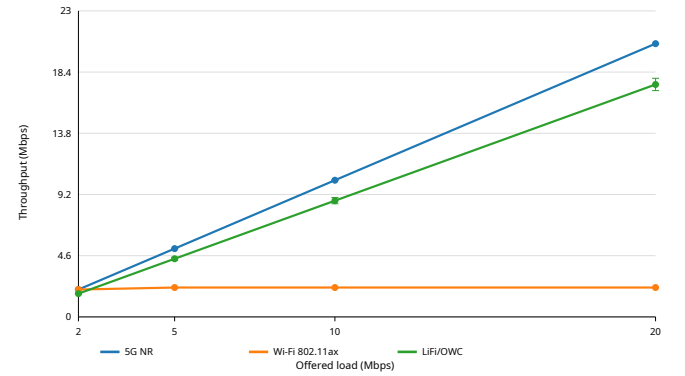


Fig. 8. Synthetic wireless throughput versus offered load. For each load and branch, the three CPE flows are averaged within each run, nuisance speed/FOV conditions are averaged within each seed, and error bars show the standard deviation across the three seeds.

and the FlowMonitor-reported packet-loss ratio increased by 0.701. These results indicate a configuration-specific queueing response in the Wi-Fi branch and load-responsive throughput in the NR and LiFi branches. Increasing mobility from 0.5 to 3 m/s changed throughput by 0 Mbps for NR, 0.00018 Mbps for Wi-Fi 802.11ax, and -2.572 Mbps for LiFi under the selected geometry and channel models. Narrowing the LiFi FOV from 70° to 40° reduced the in-FOV fraction by 0.299 and the received-packet fraction by 0.299 in matched run-level comparisons. Because the ideal concentrator gain also changes with FOV, aggregate H_{LOS} and SNR are not asserted to decrease monotonically. Branch comparisons depend on topology, channel/helper assumptions, link rates, delays, offered load, mobility, and LiFi modelling choices; they are not universal technology rankings.

Wireless Scope and Limitations. The final synthetic backend uses 5G NR via CTTC 5G-LENA, Wi-Fi 802.11ax, and an equation-driven LOS LiFi/OWC model. The LiFi model uses a stock packet-access abstraction and is not calibrated to the practical testbed. Technology comparisons are therefore specific to the declared configuration. The validation establishes implementation consistency and reproducibility of the bounded synthetic telemetry rather than physical-model fidelity.

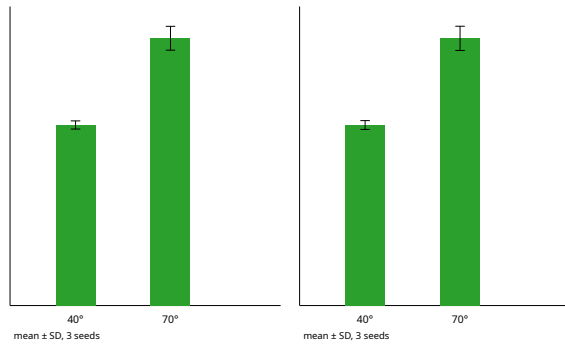


Fig. 9. Synthetic LiFi/OWC response to receiver FOV. The in-FOV fraction is computed from sampled optical states; the received-packet fraction is computed from exported LiFi flow counters. Bars show mean $\pm$ SD across three seeds after CPE and nuisance-condition averaging.

6. CONCLUSIONS

We presented an openly available dataset framework spanning independently acquired optical-transport, fibre-sensing, and wireless-access campaigns together with constructed and synthetic evaluation data. Cross-domain denotes common publication, metadata, representation, and task-level fusion; the practical campaigns are not simultaneous end-to-end measurements.

On the constructed benchmark, the three-dimensional semantic representation preserves the binary-task utility of the hand-crafted representation while moderately reducing observed CPE-attribute leakage. This is an empirical privacy-utility trade-off rather than a formal privacy guarantee. The LLM-driven multi-agent DT workflow demonstrates auditable interaction with GNPY and ns-3, while the deterministic baseline shows that exhaustive enumeration of the fixed optical space does not require an LLM. Versioned replay establishes numerical reproducibility boundaries but also exposes simulator sensitivity and residual configuration warnings. Future work will focus on calibrated physical models, contemporaneous cross-domain measurements, and repeated workflow evaluation on larger scenario spaces.

ACKNOWLEDGEMENT

This work was partly supported by the European Commission's Horizon research and innovation program: Allegro project (No. 101092766) and the EU-funded project: ECO-eNET (No. 10113933).

REFERENCES

- G. Villa, C. Tipantúa, D. S. Guamán, G. V. Arévalo, and B. Arguero, "Machine learning techniques in optical networks: A systematic mapping study," *IEEE Access* **11**, 98714–98750 (2023).
- Y. Sun, M. Peng, Y. Zhou, Y. Huang, and S. Mao, "Application of machine learning in wireless networks: Key techniques and open issues," *IEEE Commun. Surv. & Tutorials* **21**, 3072–3108 (2019).
- Z. Zhai, L. Dou, Y. He, A. P. T. Lau, and C. Xie, "Open-source data for qot estimation in optical networks from alibaba," *J. Opt. Commun. Netw.* **16**, 1–3 (2024).
- M. Filer, J. Gaudette, M. Ghobadi, R. Mahajan, T. Issenhuth, B. Klinkers, and J. Cox, "Elastic optical networking in the microsoft cloud," *J. Opt. Commun. Netw.* **8**, A45–A54 (2016).

- S. Shen, J. Han, K. Bardhi, H. Li, R. Yang, Y. Teng, V. Yokar, S. Yan, and D. Simeonidou, "Unified monitoring and telemetry platform supporting network intelligence in optical networks," *J. Opt. Commun. Netw.* **17**, 139–151 (2025).
- S. Shen, X. Zhou, W. Zhao, R. Yang, Y. Vaigai, S. Yan, and D. Simeonidou, "Open-source data for multi-domain network monitoring and sensing in optical and wireless networks," in *2026 Optical Fiber Communications Conference and Exhibition (OFC)*, (2026), pp. 1–3.
- H. He *et al.*, "Integrated sensing and communication in an optical fibre," *Light. Sci. & Appl.* **12**, 25 (2023).
- W. Zhao *et al.*, "Fibre vibration sensing and localization with cross-correlation based neural network in a practical bidirectional optical transmission system," in *Asia Communications and Photonics Conference*, (2025), pp. 1–5.
- W. Zhao *et al.*, "Train flow sensing and positioning via in-service polarization reuse over bidirectional optical fibre transmission systems," in *Optical Fiber Communication Conference*, (Optica Publishing Group, 2026), p. M4J.4.
- "Management and orchestration; 5G performance measurements," Technical Specification TS 28.552, 3GPP (2024). Available: <https://www.3gpp.org>.
- "5G; Management and orchestration; 5G end to end Key Performance Indicators (KPI)," Technical Specification TS 128 554, ETSI (2024).
- C. Delezoide, P. Ramantanis, and P. Layec, "Leveraging field data for the joint optimization of capacity and availability in low-margin optical networks," *J. Light. Technol.* **38**, 6709–6718 (2020).
- G. Bergk, B. Shariati, P. Safari, and J. K. Fischer, "ML-assisted qot estimation: A dataset collection and data visualization for dataset quality evaluation," *J. Opt. Commun. Netw.* **14**, 43–55 (2022).
- Y. He, Z. Zhai, L. Dou, L. Wang, Y. Yan, C. Xie, C. Lu, and A. P. T. Lau, "Improved qot estimations through refined signal power measurements and data-driven parameter optimizations in a disaggregated and partially loaded live production network," *J. Opt. Commun. Netw.* **15**, 638–648 (2023).
- H. Akbari, B. Shariati, J. L. Moreno Morrone, P. Safari, J. K. Fischer, and R. Freund, "Datasets for qot estimation in sdm networks," *J. Opt. Commun. Netw.* **17**, 514–525 (2025).
- Y. Li, J. Zhang, C. Yu, J. Liu, Y. Yan, and A. P. T. Lau, "Dataset for traffic monitoring using distributed acoustic sensing in busy traffic environments," in *Optical Fiber Communication Conference (OFC) 2026*, (Optica Publishing Group, 2026), Technical Digest Series, p. M3M.1.
- A. Y.-I. Yuan, C. Tan, and S. J. Savory, "Experimental semiconductor optical amplifier dataset for machine learning," in *Optical Fiber Communication Conference (OFC) 2026*, (Optica Publishing Group, 2026), Technical Digest Series, p. M3M.2.
- Q. Zhao, C. Lim, and A. Nirmalathas, "An open-source mmwave analog radio-over-fiber dataset enabling machine learning applications," in *Optical Fiber Communication Conference (OFC) 2026*, (Optica Publishing Group, 2026), Technical Digest Series, p. M3M.3.
- H. Song, T. Zhou, Z. Li, J. Shin, G. Fioccola, and Y. Lee, "Network telemetry framework," RFC 9232, Internet Engineering Task Force (2022).
- OpenConfig, "grpc network management interface specification," Tech. rep., OpenConfig (2018). OpenConfig specification.
- F. Paolucci, A. Sgambelluri, F. Cugini, and P. Castoldi, "Network telemetry streaming services in sdn-based disaggregated optical networks," *J. Light. Technol.* **36**, 3142–3149 (2018).
- T. Tanaka, T. Inui, S. Kawai, S. Kuwabara, and H. Nishizawa, "Monitoring and diagnostic technologies using deep neural networks for predictive optical network maintenance [invited]," *J. Opt. Commun. Netw.* **13**, E13–E22 (2021).
- A. Sgambelluri, A. Pacini, F. Paolucci, P. Castoldi, and L. Valcarenghi, "Reliable and scalable kafka-based framework for optical network telemetry," *J. Opt. Commun. Netw.* **13**, E42–E52 (2021).
- H. H. Mahmoud, D. Mi, A. Aneiba, M. Idrissi, G. Sharma, C. H. Foh, and P. Xiao, "A review of open-ran intelligence: Opportunities, challenges, real-life applications and impacts," *ACM Comput. Surv.* **58**,

- 1–36 (2026).
25. R. Khan, P. Kumar, D. N. K. Jayakody, and M. Liyanage, "A survey on security and privacy of 5g technologies: Potential solutions, recent advancements, and future directions," *IEEE Commun. Surv. & Tutorials* **22**, 196–248 (2020).
26. H. Xie, Z. Qin, X. Tao, and K. B. Letaief, "Task-oriented multi-user semantic communications," *IEEE J. on Sel. Areas Commun.* **40**, 2584–2597 (2022).
27. S. Ma, W. Qiao, Y. Wu, H. Li, G. Shi, D. Gao, Y. Shi, S. Li, and N. Al-Dhahir, "Task-oriented explainable semantic communications," *IEEE Transactions on Wirel. Commun.* **22**, 9248–9262 (2023).
28. A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy, "Deep variational information bottleneck," in *International Conference on Learning Representations*, (2017).
29. H. Edwards and A. Storkey, "Censoring representations with an adversary," in *International Conference on Learning Representations*, (2016).
30. D. Madras, E. Creager, T. Pitassi, and R. Zemel, "Learning adversarially fair and transferable representations," in *Proceedings of the 35th International Conference on Machine Learning*, vol. 80 of *Proceedings of Machine Learning Research* (2018), pp. 3384–3393.
31. Y. Wang, S. Guo, Y. Deng, H. Zhang, and Y. Fang, "Privacy-preserving task-oriented semantic communications against model inversion attacks," *IEEE Transactions on Wirel. Commun.* **23**, 10150–10165 (2024).
32. Q. Zhuge, X. Liu, Y. Zhang, M. Cai, Y. Liu, Q. Qiu, X. Zhong, J. Wu, R. Gao, L. Yi, and W. Hu, "Building a digital twin for intelligent optical networks," *J. Opt. Commun. Netw.* **15**, C242–C262 (2023).
33. V. Curri *et al.*, "Gnpy model of the physical layer for open and disaggregated optical networking," *J. Opt. Commun. Netw.* **14**, C92–C104 (2022).
34. M. C. Mohamed, R. Ambrosone, R. D'Ingillo, M. U. Masood, G. Malik, S. Straullu, S. K. Bhyri, J. Pedro, A. Napoli, G. M. Galimberti, W. Wakim, and V. Curri, "Leveraging digital twins to build comprehensive network failure databases for predictive machine learning in optical transport networks," in *2025 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, (2025), pp. 1–6.
35. M. C. Mohamed, M. U. Masood, R. Ambrosone, G. Malik, R. D'Ingillo, S. Straullu, S. K. Bhyri, G. M. Galimberti, J. Pedro, A. Napoli, W. Wakim, and V. Curri, "Machine learning agents leveraging digital twins for failure prediction in optical networks," in *2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, (2025), pp. 1–6.
36. G. F. Riley and T. R. Henderson, "The ns-3 network simulator," in *Modeling and Tools for Network Simulation*, K. Wehrle, M. Güneş, and J. Gross, eds. (Springer, 2010), pp. 15–34.
37. M. Polese, L. Bonati, S. D'Oro, P. Johari, D. Villa, S. Velumani, R. Gangula, M. Tsampazi, C. P. Robinson, G. Gemmi, A. Lacava, S. Maxenti, H. Cheng, and T. Melodia, "Colosseum: The open ran digital twin," *IEEE Open J. Commun. Soc.* **5**, 5452–5466 (2024).
38. Z. Tao, W. Xu, Y. Huang, X. Wang, and X. You, "Wireless network digital twin for 6g: Generative ai as a key enabler," *IEEE Wirel. Commun.* **31**, 24–31 (2024).
39. H. Chai, H. Wang, T. Li, and Z. Wang, "Generative ai-driven digital twin for mobile networks," *IEEE Netw.* **38**, 84–92 (2024).
40. D. Wang, Y. Song, Y. Zhang, X. Jiang, J. Dong, F. N. Khan, T. Sasai, S. Huang, A. P. T. Lau, M. Tornatore, and M. Zhang, "Digital twin of optical networks: A review of recent advances and future trends," *J. Light. Technol.* **42**, 4233–4259 (2024).
41. Y. Zhang, M. Zhang, Y. Song, X. Jiang, Y. Wang, S. Shen, and D. Wang, "Gpt-enabled digital twin assistant for multi-task cooperative management in autonomous optical network," in *Optical Fiber Communication Conference (OFC) 2024*, (Optica Publishing Group, 2024), Technical Digest Series, p. Th1G.4.
42. Y. Song, Y. Zhang, A. Zhou, Y. Shi, S. Shen, X. Tang, J. Li, M. Zhang, and D. Wang, "Synergistic interplay of large language model and digital twin for autonomous optical networks: Field demonstrations," *IEEE Commun. Mag.* **63**, 90–96 (2025).
43. C. Sun, X. Yang, N. Di Cicco, R. Ayassi, V. V. Garbhapu, P. A. Stavrou, M. Tornatore, G. Charlet, and Y. Pointurier, "Experimental demonstration of local ai-agents for lifecycle management and control automation of optical networks," *J. Opt. Commun. Netw.* **17**, C82–C92 (2025).
44. X. Liu, Q. Qiu, Y. Zhang, M. Cai, Y. Liu, L. Yi, W. Hu, and Q. Zhuge, "Auto-dtwave: Digital twin-aided autonomous optical network operation with continuous wavelength loading," in *Optical Fiber Communication Conference (OFC) 2024*, (Optica Publishing Group, 2024), Technical Digest Series, p. Th1G.5.
45. Q. Zhuge, X. Liu, Y. Zhang, and Q. Qiu, "Digital twin and ai agent for autonomous optical networks," in *Optical Fiber Communication Conference (OFC) 2025*, (Optica Publishing Group, 2025), Technical Digest Series, p. W4G.1.
46. X. Liu, Y. Zhang, Q. Qiu, Y. Cheng, W. Hu, and Q. Zhuge, "Field trial of an llm-powered ai agent for autonomous optical networks: A full-lifecycle demonstration," *J. Opt. Commun. Netw.* **18**, A179–A190 (2026).
47. Smart Internet Lab, University of Bristol, "Practical Datasets," GitHub repository (2025). [Online]. Available: <https://github.com/hpn-bristol/Open-Source-Data-for-Multi-Domain-Network-Monitoring-and-Sensing-in-Optical-and-Wireless-Networks>.
48. Smart Internet Lab, University of Bristol, "Final W4C Three-RAT Synthetic Wireless Dataset and Revision Artifacts," GitHub repository (2026). [Online]. Available: https://github.com/WanxinZhao/JOCN-Dataset-WanxinCopy-/tree/main/W4C_THREE_RAT_SWEEP/; source code, configuration, and run-level data for the final three-RAT synthetic wireless experiment.
49. R. Yang, S. Shen, H. Li, Z. Shi, R. Wang, R. Nejabati, S. Yan, and D. Simeonidou, "986 km field trial of cascaded ann-based link-penalty models for qot prediction," in *2023 Optical Fiber Communications Conference and Exhibition (OFC)*, (2023), pp. 1–3.
50. W. Zhao, "JOCN Dataset Code and Major-Revision Reproduction Artefacts," GitHub repository, commit 82914127b70acf3b220bc09a56a179cb7abd1252 (2026). [Online]. Available: <https://github.com/WanxinZhao/JOCN-DATASET-Code/tree/82914127b70acf3b220bc09a56a179cb7abd1252>.
51. J. M. Kahn and J. R. Barry, "Wireless infrared communications," *Proc. IEEE* **85**, 265–298 (1997).
52. T. Komine and M. Nakagawa, "Fundamental analysis for visible-light communication system using led lights," *IEEE Transactions on Consumer Electron.* **50**, 100–107 (2004).